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BE (6th Semester)

Examination, April - May, 2016

(New Scheme)

Computer Graphics and Animation

Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 28

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Note : (*i*) Part (*a*) of each unit is compulsory. Attempt any two parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

Unit-I

1. (*a*) What is the difference between raster and random scan? 2

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(*b*) Explain DDA algorithm and also draw line from (0, 0) to (6, 6) using this algorithm. 7

(*c*) Determine positions along the circle octant in the first quadrant from $x = 0$ to $y = 0$ where radius $r = 5$. 7

(*d*) Explain boundary fill and flood fill algorithm and also write their procedures. 7

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Unit-II

2. (*a*) Find scaling factor 5x and 5y where the lower left corner of window is (1, 1) and upper right corner is at (3, 5) and lower left corner of view port is (0, 0) and upper right corner of view port is (1, 1). 2

(*b*) Explain in brief about S D Cohen-Sutherland line clipping algorithm. 7

(*c*) A line from P_0 (1, –1, –1) to P_1 (0.5, 3, 1) is to be clipped against the clipping volume defined by a cube with two diagonally opposite corners at (0, 0, 0) and (2, 2, 2). Use Cohen-Sutherland line clipping algorithm. 7

(*d*) Explain in brief about Cyrus Beck line clipping algorithm. 7

Unit-III

3. (*a*) Determine the equation of a cubic spline segment passing through two points $A(0, 5)$ and $B(10, 20)$ and having a slope of 2 and curvature of –0.5 at A. 2

(*b*) What are the important properties of spline curve? 7

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(*c*) A cubic Bezier curve segment is described by the control points $P_0(20, 20)$, $P_1(40, 80)$, P_2 (80, 80), P_3 (90, 50). Another curve segment is described by $Q_0(a, b)$, $Q_1(c, 20)$, $Q_2(150, 20)$, and $Q_3(180, 20)$. 7

Determine the value of a, b, c, so that the two curve segments should be smoothly joined.

(*d*) For $n = d = 3$ calculate knot value and B spline blending function. 7

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Unit-IV

4. (*a*) What are the effects of perspective projection? 2

(*b*) Find the perspective projection of an edge having two end points $A(-1, -1, -2d)$ and $B(2, 2, 0)$ on the projection plane where $Z' = 0$ given that the COP is at $(0, 0, -d)$. 7

(*c*) Explain in brief about 2 buffer algorithm. 7

(*d*) What is the importance of ambient light? Also explain difference between diffuse and specular reflection.

Unit-V

5. (*a*) What are self similar fractals? 2

(*b*) What are the important steps in the design of animation sequence? 7

(*c*) What is the importance of volume rendering? 7

(*d*) Write short notes on: 7

(*i*) Texture mapping

(*ii*) Bump mapping

(*iii*) Phong shading

(*iv*) Graud shading

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